

Chaitanyaa Chopraa

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Summary

Computer Science graduate (UVic, Dec 2025) with hands-on experience building interactive systems in Unity and Unreal. Strong foundation in C#, gameplay architecture, performance optimization, and cross-platform design. Experienced collaborating in Agile environments, using Git workflows, and delivering production-quality features with a focus on user experience, maintainability, and scalability.

Education

University of Victoria

BSc, Computer Science — Software Systems • Business Minor

Victoria, BC, Canada

Sept 2021 – Dec 2025

Graduating GPA: 8.3 / 9.0

Relevant Coursework: Software Engineering, Web Development, Databases, Algorithms and Data Structures, Operating Systems, Software Testing and QA

Skills

Game Development: Unity, C#, Gameplay Systems, Physics, Input Systems, UI, Animation, Procedural Generation

Client & Platforms: Mobile fundamentals, WebGL concepts, Performance Optimization, Profiling, Memory Awareness

Software Engineering: OOP, Design Patterns, Architecture, Debugging, Testing, Code Reviews

Tools: Git, Docker, Linux, JMeter, CI/CD Concepts

Collaboration: Agile/Scrum, Sprint Planning, Documentation, QA Communication

Technical Projects

Procedural Building Generator

Unity, C#, Procedural Generation

- Developed a **procedural building generation system** in Unity using C#, enabling automatic creation of modular and scalable building layouts.
- Designed **L-systems algorithms** to dynamically assemble structures from reusable components, increasing level variety and design efficiency.
- Focused on **performance optimization and editor usability** to support rapid iteration and real-time generation.

Mini Football Game

Unreal Engine, Blueprints, Game Physics

- Built a **3D mini football** game prototype in **Unreal Engine**, implementing player controls, ball physics, goal detection, and scoring logic.
- Designed interactive gameplay mechanics using **Unreal Engine Blueprints** to ensure responsive movement and realistic ball interaction.
- Implemented **visual and gameplay feedback systems**, including score updates and goal reactions, to enhance player experience.
- Added **multiplayer functionality** using replication to synchronize gameplay and game state across clients.

Social Spark — AI Brand & Influencer Platform

React, Node.js, MongoDB, Gemini API, Docker

- Developed an **AI-driven platform** to improve brand–influencer collaboration through intelligent campaign matching and analytics.
- Integrated the **Gemini API** to automate content evaluation, generate campaign recommendations, and provide engagement insights.
- Designed **secure authentication and backend services** to support scalable user interactions and data processing.
- Designed **scalable backend services** with clear API boundaries to support future feature expansion.

Professional Experience

Houle Electric

Jan 2024 – Aug 2024

Co-op Project Manager / Assistant

Cowichan District Hospital Redevelopment

- Developed and implemented robust **Quality Assurance (QA) and Quality Control (QC)** processes for the Cowichan District Hospital redevelopment project.
- Successfully executed effective QA/QC workflows to ensure project completion on time and in compliance with quality, safety, and regulatory standards.
- Played a pivotal role in **configuring Procore** by designing a reusable, company-wide QA/QC template, now actively adopted across multiple projects.
- Managed high-volume documentation, inspections, and reporting, maintaining accuracy and accountability across field and management teams.

Infinity Outdoor Pvt. Ltd

Delivered 2025

Contract Web Developer

- Designed and launched a **responsive marketing** website to enhance brand visibility and showcase advertising campaigns.
- Built the application using **React, Vite, and Tailwind CSS**, achieving a 35% improvement in load times through optimized component architecture.
- Collaborated closely with stakeholders to align design, content, and functionality, delivering a modern and data-driven UX.